

Towards comparing the two approaches to CLC

Ian Guleason

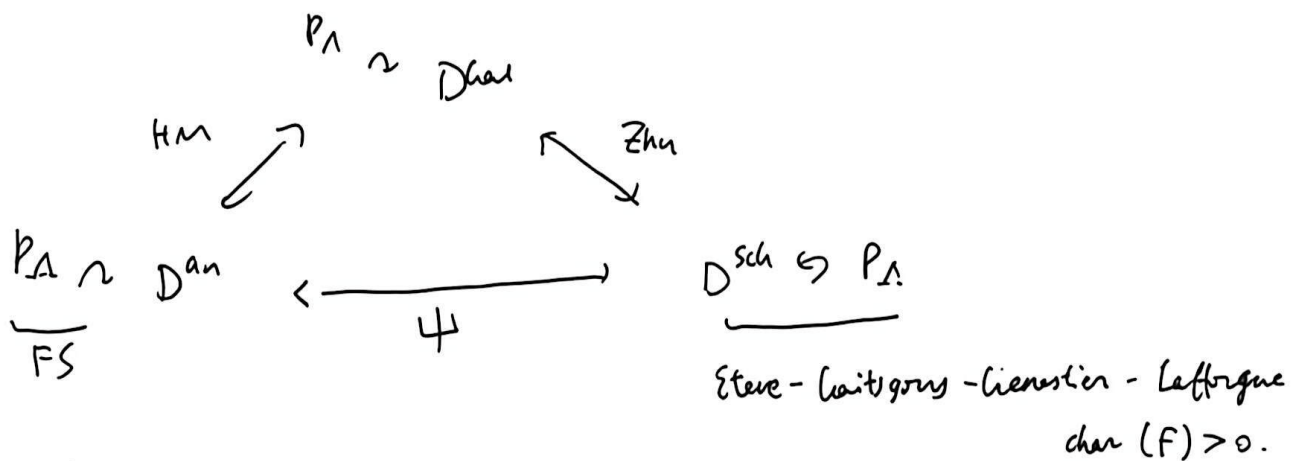
Lecture 1. Notation: $p \neq \ell$ primes, F p -adic field, G q -split reductive / F

I Iwahori / $\mathcal{O}_F$. Λ $\mathbb{Z}_\ell$ -alg (ℓ -torsion)

$P_\Lambda = \text{Perf}(P_{\text{an}})$, $D^{\text{Gal}} = \text{Ind}_{\text{Gal}}^{\text{un}}(P_{\text{an}})$.

$D^{\text{an}} = D_{\text{ét}}^{\text{an}}(\text{Bun}_G)$, $D^{\text{sch}} = \text{Shv}(\text{Isoc})$

Ideal picture:



Why care about Ψ ?

1) D^{an}

gen. fibre of
Sh. var

p -adic period domain

D^{sch}

special fibre of
Sh. var.

ADLV

2) Different tools:

D^{an} - Eisenstein series

D^{sch} - categorical trace technique

Th (G. - Hamann - Ivanov - Lorenzo - Zou) If Λ is a torsion $\mathbb{Z}_\ell$ -alg, then

we can exhibit a can. equiv. in $\text{Lin}(\text{at}_\Lambda$: $\Psi: \text{Shv}(\text{Isoc}_G, \Lambda) \xrightarrow{\sim} D_{\text{ét}}^{\text{an}}(\text{Bun}_G, \Lambda)$.

Moreover, Ψ is t-exact.

$$- \psi(P_{D_{\leq 0}^{sch}}, P_{D_{\geq 0}^{sch}}) = (h_{D_{\leq 0}^{an}}, h_{D_{\geq 0}^{an}})$$

$$- \psi(e_{D_{\leq 0}^{sch}}, e_{D_{\geq 0}^{sch}}) = (p_{D_{\leq 0}^{an}}, p_{D_{\geq 0}^{an}})$$

Naive construction of ψ .

$X \in \text{PreStk} \rightsquigarrow V\text{-stack}$

$$X^\diamond(R, R^+) = X(\text{Spec } R) \quad + \text{ sheafity.}$$

$$\text{Scholze's analytification} : c^* : \text{Det}(X, \Lambda) \rightarrow \text{Det}(X^\diamond, \Lambda)$$

$$\text{Bun}_G^{\text{mer}} \xrightarrow{\sigma} \text{Bun}_G$$

$$\gamma \downarrow \\ \text{Isoc}_G^\diamond$$

$$\text{Bun}_G^{(R, R^+)} = G\text{-bundles on } X_{(0, \infty)}^R + \text{Frob}$$

$$\text{Isoc}_G^{\diamond(R, R^+)} = G\text{-bundles on } \text{Spec } W(R) \left[\frac{1}{\pi}\right] + \text{Frob}$$

$$Y_{(0, \infty)}^R \supset Y_{(0, \infty)}^R$$

$$Y_{(0, \infty)}^R \left[\frac{1}{\pi}\right] \supset Y_{(0, \infty)}^R$$

$$\uparrow \\ \text{Spec } W(R) = Y_{(0, \infty)}^R \wedge_{\pi=0}$$

$$\cup \\ \text{Spec } W(R) \left[\frac{1}{\pi}\right]$$

$$\psi^{\text{naive}} = \sigma! \gamma^* c^* : \text{Det}(\text{Isoc}_G) \rightarrow \text{Det}(\text{Isoc}_G^\diamond) \rightarrow \text{Det}(\text{Bun}_G)$$

Technical problems, a) Does $\sigma!$ exist?

$$b) \text{Shv}(\text{Isoc}_G) = \text{Det}(\text{Isoc}_G) ?$$

Solution, Treat $\text{Bun}_G^{\text{mer}}$ as a sind-Artin V -stack.
(sifted + ind)
 $\longrightarrow$

Étale sheaves on schemes

$$\left\{ \text{ptp schemes} / \overline{\mathbb{F}_p} \right\} \subset \left\{ \text{qcqs schemes} / \overline{\mathbb{F}_p} \right\} \subset \left\{ \text{étale stacks} \right\}$$

a) Dét (for which c^* is defined)

b) Sh_v^* (intermediate)

c) $\text{Sh}_v = \text{Sh}_v^!$ (last week)

- All agree on pt-p - schemes.

- On qcqs schemes: $\text{Sh}_v^*(X) = \text{LK}_{f^*} \text{Sh}_v^*(X_i)$

$$\text{Sh}_v^!(X) = \text{LK}_{f^!} \text{Sh}_v^!(X_i)$$

$$\text{Dét}(X) = \text{left-completion}(\text{Sh}_v^*)$$

$$\text{Sh}_v^* \Rightarrow \text{Dét}$$

- On étale stacks: $\text{Sh}_v^! = \text{RK}_{f^!} \text{Sh}_v^!$

$$\text{Sh}_v^* = \text{RK}_{f^*} \text{Sh}_v^*$$

$$\text{Dét} = \text{RK}_{f^*} \text{Dét}$$

= "finite" hierarchy

$$\left\{ \begin{array}{c} \text{pt-p} \\ \text{Alg Sp} \end{array} \right\} \subset \left\{ \begin{array}{c} \text{standard} \\ \text{placid Alg Sp} \end{array} \right\} \subset \left\{ \begin{array}{c} \text{very} \\ \text{placid} \\ \text{stacks} \end{array} \right\} \subset \left\{ \begin{array}{c} \text{ind} \\ \text{vers} \\ \text{placid} \end{array} \right\} \subset \left\{ \begin{array}{c} \text{sing} \\ \text{very} \\ \text{placid} \end{array} \right\}$$

$\downarrow$

$$\text{Pl} \leq \mu \leftarrow \text{Lg} \leq \mu \rightarrow \text{Sh} \leq \mu$$

$$\text{Sh} \leq \mu$$

$\downarrow$

$$\text{Sh} \leq \mu \xrightarrow{\text{Nt}} \text{Isoc}_G$$

$\parallel$

$$\text{colim}_{\mu \in X_G(T)} \text{Sh}_Z^{\leq \mu}$$

$$\text{Sh}^{\leq \mu} = \frac{\text{Lg} \leq \mu}{\text{Adp } L^+ I}$$

$$\underbrace{\text{Fe}}_{\text{wip.}} \rightarrow \underbrace{L^+ I}_{\text{findin}} \rightarrow \mu \rightarrow \dots$$

Def. If X is a $\text{ind-very placid stack}$, we say $f: Z \rightarrow X$ is a bounded piece if Z is very placid and f is ind-plt proper.

$$\underline{B}X \subset \{ \text{étale stacks} / X \}$$

Fact. If X is ind-very placid , $\text{Shv}^!(X)^\vee \simeq \varinjlim_{\substack{Y \in B_X \\ t! = f^*}} \text{Shv}^*(Y)$

Def. Given $Y \in B_{\text{Isoc}_G}$, we let $Y_Y = Y \times_{\text{Isoc}_G^{\text{ét}}} \text{Bun}_G^{\text{mor}}$

we let $\sigma_Y: Y_Y \rightarrow \text{Bun}_G^{\text{mor}} \rightarrow \text{Bun}_G$

We say that Y_Y is σ -compatible if σ_Y is $!$ -able. $B_{\text{Isoc}_G}^{\sigma} \subset B_{\text{Isoc}_G}$

Th (G.-Ivanov-Zilber)

$$\left(\text{Sht}_{\mathbb{I}}^{G, \leq w} \right)_Y = \text{Sht}_{\mathbb{I}, \mathbb{F}_p}^{G, \leq w}$$

Th. The map $\text{Sht}_{\mathbb{I}, \mathbb{F}_p}^{G, \leq w} \rightarrow \text{Bun}_G$ is representable in locally spatial diamonds and $!$ -able.

Lem. $\text{Sht}_{\mathbb{I}}^{G, \leq w} \in B_{\text{Isoc}_G}^{\sigma}$, $\forall w \in W$, and $\text{Shv}(\text{Isoc}_G)^\vee \simeq \varinjlim_{Y \in B_{\text{Isoc}_G}^{\sigma}} \text{Shv}^*(Y)$.

— Σ

Th (Zhu) $\text{ID}^{\text{can}}: \text{Shv}^!(\text{Isoc}_G) \simeq \text{Shv}^!(\text{Isoc}_G)^\vee$

$$\psi := \psi^\vee \circ \text{ID}^{\text{can}}, \quad \psi^\vee := \varinjlim_{Y \in B_{\text{Isoc}_G}^{\sigma}} \psi_Y$$

$$\psi_Y: \text{Shv}^*(Y) \xrightarrow{\hat{c}} \text{Dét}(Y) \xrightarrow{c^*} \text{Dét}(Y^0) \xrightarrow{\gamma_Y^*} \text{Dét}(Y_Y) \xrightarrow{\sigma_Y!} \text{Dét}(\text{Bun}_G)$$

Petersen

Technical point: Ψ_Y should be functorial in B_{Isoc_G} .

- $(\hat{})$, γ^* , σ_Y ! easy

- c^* requires more work

Def. An étale stack is resilient if $X^{\diamond_{\text{proét}}} \rightarrow X^{\diamond_{\text{ét}}}$ is an equiv.

A map of étale stacks $X_1 \rightarrow X_2$ is Artin if there is $Y \rightarrow X_1$ coh, smooth, representable ptp, surjective, s.t. $Y \rightarrow X_2$ is also representable and ptp.

Th (GILZ) There is a canonical natl transf of $\mathfrak{f}$ -functor formalism

$$\text{Dét}(-, \Lambda) \Rightarrow \text{Dét}(-^{\diamond}, \Lambda) \text{ defined on } (\mathcal{C}, \mathcal{E}).$$

$$\mathcal{E} \in \{E_1, E_2\} : \quad \mathcal{C} = \left\{ \begin{array}{l} \text{resilient} \\ \text{étale stacks} \end{array} \right\}, \quad E_1 = \{ \text{"analytifiable maps"} \}$$

$$E_2 = \{ \text{"Artin maps"} \}$$

Lecture 2. $\text{Convex}_{B(G)} = \left\{ V \subset B(G) : \begin{array}{l} V \text{ is finite} \\ [a, b] \subset V, \forall a, b \in V \end{array} \right\}$

$$V_1 \subset V_2 \quad \rightsquigarrow \quad \text{Isoc}_G^{V_1} \xrightarrow{i_{V_1, V_2}} \text{Isoc}_G^{V_2}$$

$$\left. \begin{array}{c} \downarrow \\ \text{Bun}_G^{V_1} \hookrightarrow \text{Bun}_G^{V_2} \\ \downarrow j_{V_1, V_2} \end{array} \right\}$$

(Th.)

Ψ intertwines the following operations:

$$\underline{\text{Isoc}}: \quad i_! \rightarrow i^! \rightarrow i_b \quad \quad i^* \rightarrow i_* \rightarrow i^\#$$

$$\underline{\text{Bun}}_G: \quad j_\# \rightarrow j^* \rightarrow j_x \quad \quad j^b \rightarrow j_! \rightarrow j^!$$

We have $B(G)$ -indexed SOD

$$\mathcal{S}_{\text{Isoc}_G} : \text{Conv}(\text{Convex}) \rightarrow \text{LinCat}_\Lambda$$

$$\mathcal{S}_{\text{Isoc}_G}(V) := \text{Sh}_V^!(\text{Isoc}_G^V)$$

$$\mathcal{S} \left(\begin{array}{ccc} & \xleftarrow{w} & \\ v & & u \end{array} \right) := i_{wU}^! \circ j_{wV}^*$$

$$\mathcal{S}_{\text{Bun}_G}^{\text{ex}} : \text{Conv}(\text{Convex}) \rightarrow \text{LinCat}_\Lambda$$

$$\mathcal{S}_{\text{Bun}_G}^{\text{ex}}(V) = \text{Det}(\text{Bun}_G^V)$$

$$\mathcal{S}_{\text{Bun}_G}^{\text{ex}} \left(\begin{array}{ccc} & \xleftarrow{w} & \\ v & & u \end{array} \right) := \hat{j}_{wU} \# \circ \hat{j}_{wV}^b.$$

$$\hat{j}_{wU} \# := \text{ID}_{BZ} \circ j_{wU}^! \circ \text{ID}_{BZ}$$

$$\hat{j}_{wV}^b := \text{ID}_{BZ} \circ j_{wV}^* \circ \text{ID}_{BZ}$$

Prop ψ promotes to an equiv. of functors $\psi_{\mathcal{S}} : \mathcal{S}_{\text{Isoc}_G} \xrightarrow{\sim} \mathcal{S}_{\text{Bun}_G}^{\text{ex}}$

$$V \in \text{Convex}, \quad \psi_V : \text{Sh}_V^!(\text{Isoc}_G^V) \xrightarrow{\sim} \text{Det}(\text{Bun}_G^V)$$

$$\{b\} \in \text{Convex}, \quad \psi_b : \text{Sh}_V^!(\text{Isoc}_G^b) \xrightarrow{\sim} \text{Det}(\text{Bun}_G^b)$$

Criterion: ψ is an equiv. if

$$a) \psi \left(i_b^! \left(\text{Sh}_V^!(\text{Isoc}_G^b) \right) \right) \subset \hat{j}_b \# \left(\text{Det}(\text{Bun}_G^b) \right)$$

$$b) \forall V \in \text{Convex} \text{ and } b \in V \text{ minimal (closed)}$$

$$\psi_V \left(i_{V \setminus \{b\} V}^! \left(\text{Sh}_V(\text{Isoc}_G^{V \setminus \{b\}}) \right) \right) \subset \hat{j}_{V \setminus \{b\} V}^! \left(\text{Det}(\text{Bun}_G^{V \setminus \{b\}}) \right)$$

$$c) \psi_b : \text{Sh}_V(\text{Isoc}_G^b) \xrightarrow{\sim} \text{Det}(\text{Bun}_G^b) \text{ is an equiv.}$$

Th (G. - Imamura - Zimmig) We have a diagram w/ Cartesian square:

$$\begin{array}{ccccc}
 & & \pi_b & & \\
 & & \curvearrowright & & \\
 \coprod_{b \in B(a)} \mu_b & \xrightarrow{k_b} & \text{Bun}_a^{\text{mer}} & \xrightarrow{\sigma} & \text{Bun}_a \\
 \downarrow q_b & & \downarrow \gamma & & \\
 \coprod_{b \in B(a)} (\text{Isoc}_a^b)^\diamond & \longrightarrow & \text{Isoc}_a^\diamond & &
 \end{array}$$

$$\sim \mu_b = (\text{Bun}_a^{\text{mer}})^{r=b}$$

$$\sim \tau_b = (\text{Bun}_a^{\text{mer}})^{b=b}$$

$$\begin{aligned}
 \psi^{\text{naive}}(i_b!(-)) &\cong \sigma! \gamma^* c^* i_b! \\
 &\cong \sigma! \gamma^* i_b! c^* \\
 &= \sigma! k_b! q_b^* c^* \\
 &\cong \pi_b! q_b^* c^*
 \end{aligned}$$

$$\subset j_b^* (D(\text{Bun}_a^b))$$

Non-naive version:

$$\begin{array}{ccccccc}
 \tilde{M}_b / I_w_b & \xrightarrow{\text{open}} & \text{Sh}_I^{a, s_w_b} & \xrightarrow{\text{Sh}_I} & \text{Bun}_a^{\text{mer}} \\
 \downarrow & & \downarrow & & \downarrow \\
 \text{Sh}_I^{w_b} & \hookrightarrow & \text{Sh}_I^{s_w_b \diamond} & \xrightarrow{\text{Sh}_I^\diamond} & \text{Isoc}_a^\diamond \\
 [x/I_w_b] & & \text{Ntw}_b & &
 \end{array}$$

$$B(a) = \text{BW}_{\text{str}} \subset \frac{w}{\text{Ad} \varphi w}$$

$$I_w_b \subset G_b(F)$$

$i_b! (\text{Sh}_I(\text{Isoc}_a^b))$ is gen. under colimits by

$$\text{Ntw}_b! (\text{Sh}_I(\text{Sh}_I^{w_b}))$$

Formula for ψ_v :

$$\text{Bun}_a^{\text{mer}, v} \xrightarrow{\sigma_v} \text{Bun}_a^v$$

$$\gamma_v \downarrow \\ \text{Isoc}_a^v$$

$$\text{Bun}_a^{\text{mer}, v} = \text{Bun}_a^{\text{mer}} \times_{(\text{Bun}_a \times \text{Isoc}_a^\diamond)} (\text{Bun}_a^v \times \text{Isoc}_a^\diamond)$$

$$\psi_v^{\text{naive}} = \sigma_v! \gamma_v^* c^*$$

$$c) \left(\text{Bun}_G^{\text{naive}} \right)^{r=r=0} = \left(* / \mathcal{U}_G(F) \right) \rightarrow \text{Bun}_G^b$$

$$\downarrow$$

$$\text{Isoc}_G^b = \left[* / \mathcal{U}_G(F) \right]$$

Hardest step b)

$\forall V \in \text{Conex}$ and $b \in V$ minimal, $U = V \setminus \{b\}$

$$\psi_V \left(i_{U,*} \text{Sh}_V(\text{Isoc}_G^U) \right) \subset j_{U,!} \left(D(\text{Bun}_G^U) \right)$$

$$\beta \in U, \quad \text{Im} \left(\psi_V^{\text{naive}} i_{\beta,*} \right) \subset j_{U,!} \left(D(\text{Bun}_G^U) \right)$$

$$\Leftrightarrow j_{b,!}^* \psi_V^{\text{naive}} i_{\beta,*} = 0$$

$$V = U \cup \{b\}, \quad b \in V \text{ minimal}$$

$$\psi_V i_{\beta,*} \simeq \sigma_{V,!} \gamma_V^* c^* i_{\beta V,*}$$

$$\simeq \sigma_{V,!} \gamma_V^* i_{\beta V,*}^0 c^*$$

ULA trick

$$\simeq \sigma_{V,!} \kappa_{\beta,*} q_{\beta}^* c^*$$

pro-unipotent

base change

$$j_{b,!}^* \psi_V^{\text{naive}} i_{\beta,*} \simeq j_{U,!}^* \sigma_{V,!} \kappa_{\beta,*} q_{\beta}^* c^*$$

$$\simeq \sigma_{b,!} \kappa_{\beta}^b * \gamma_{\beta}^b c^*$$

$$\xrightarrow{\text{homogeneity}} \simeq 0$$

ULA trick.

Prop 4.18

If $f: X \rightarrow Y$ is a map of perfect art. étale stacks that is ptp reptile

and $A \in D_{\text{cons}}(X)$, then $c^* f^* A \simeq f_*^0 c^* A$.

Key input: If $X \rightarrow *$ is ptp, then $A \in D_{\text{cons}}(X) \Leftrightarrow A$ is square (ULA).

Pro-unip. BC

$$Y \cong X/H$$

$$e \rightarrow \underbrace{H_u}_{\text{pro-unip}} \rightarrow H \rightarrow H' \rightarrow e$$

$\underbrace{\hspace{1.5cm}}_{\text{f'n. dim.}}$

placid geometry: $X \rightarrow Y$ "smooth enough"

$$\begin{array}{ccc} X_u & \rightarrow & X \\ t_u \downarrow & & \downarrow t \\ U & \xrightarrow{j} & Y \end{array}$$

$$f^* j_* \simeq j_* * t_u^*$$

Prove a version $|A'|$ is played by $\overline{|B'|}$ (can. compactification of closed unit ball)

Th (G.-Iannucci - Zimmer) $\text{Sht}_I^{\text{an}} \simeq \frac{L_G^+}{\text{Ad}_\varphi(L_{t_u})^+}$

$$X \text{ (pre-stk)} \rightarrow X^+(\mathbb{R}, \mathbb{R}^+) \simeq X(\text{Spec } \mathbb{R}^0)$$

$$(|A'|)^+ = \overline{|B'|}$$